EV SALES FORECASTING & MARKET ANALYSIS

Data Analytics & Machine Learning Project

Tools: Python | SQL | Scikit-Learn | Pandas

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BUSINESS PROBLEM

THE ELECTRIC VEHICLE (EV) MARKET IN INDIA IS GROWING RAPIDLY, BUT THE GROWTH IS UNEVEN ACROSS STATES AND VEHICLE CATEGORIES. WITHOUT PROPER ANALYSIS:

- DEMAND PATTERNS REMAIN UNCLEAR
- INFRASTRUCTURE PLANNING BECOMES INEFFICIENT
- MARKET OPPORTUNITIES ARE MISSED

PROBLEM STATEMENT:

- UNDERSTAND EV ADOPTION TRENDS ACROSS INDIA
- IDENTIFY KEY GROWTH REGIONS
- PREDICT FUTURE EV DEMAND USING DATA

PROJECT OBJECTIVES

- ANALYZE EV SALES TRENDS ACROSS YEARS AND STATES
- IDENTIFY TOP-PERFORMING REGIONS AND SEGMENTS
- PERFORM CATEGORY-WISE AND TYPE-WISE ANALYSIS
- BUILD MACHINE LEARNING MODEL FOR FORECASTING
- PROVIDE ACTIONABLE BUSINESS INSIGHTS

DATASET OVERVIEW

DATASET SIZE

- EV SALES DATA FROM JAN 2014 to DEC2024
- DATASET CONTAINS ~96,000 EV SALES RECORDS
- COVERS ALL INDIAN STATES

KEY FEATURES:

- STATE
- VEHICLE CATEGORY
- VEHICLE TYPE
- EV SALES QUANTITY
-

BUSINESS VALUE:

- HELPS ANALYZE MARKET TRENDS
- SUPPORTS FORECASTING
- ENABLES STRATEGIC DECISION-MAKING

DATA ANALYTICS ARCHITECTURE

1. **DATA COLLECTION**
2. **DATA CLEANING**
3. **FEATURE ENGINEERING**
4. **EVALUATION**
5. **MODEL TRAINING**
6. **FORECASTING**

DATA PREPROCESSING

- CONVERTED DATE COLUMN INTO DATETIME FORMAT
- REMOVED INCONSISTENCIES AND HANDLED MISSING VALUES
- STANDARDIZED CATEGORICAL VARIABLES
- OPTIMIZED DATASET FOR ANALYSIS

WHY THIS IS IMPORTANT??

- CLEAN DATA IMPROVES MODEL ACCURACY AND ENSURES RELIABLE INSIGHTS

FEATURE ENGINEERING

EXTRACTED TIME-BASED FEATURES

- MONTH → CAPTURES SEASONAL TRENDS
- DAY → CAPTURES DAILY VARIATION
- YEAR → CAPTURES GROWTH PATTERN

APPLIED ENCODING ON CATEGORICAL VARIABLES

IMPACT

- CAPTURES SEASONAL TRENDS
- IMPROVES PREDICTION ACCURACY
- ENHANCES MODEL LEARNING

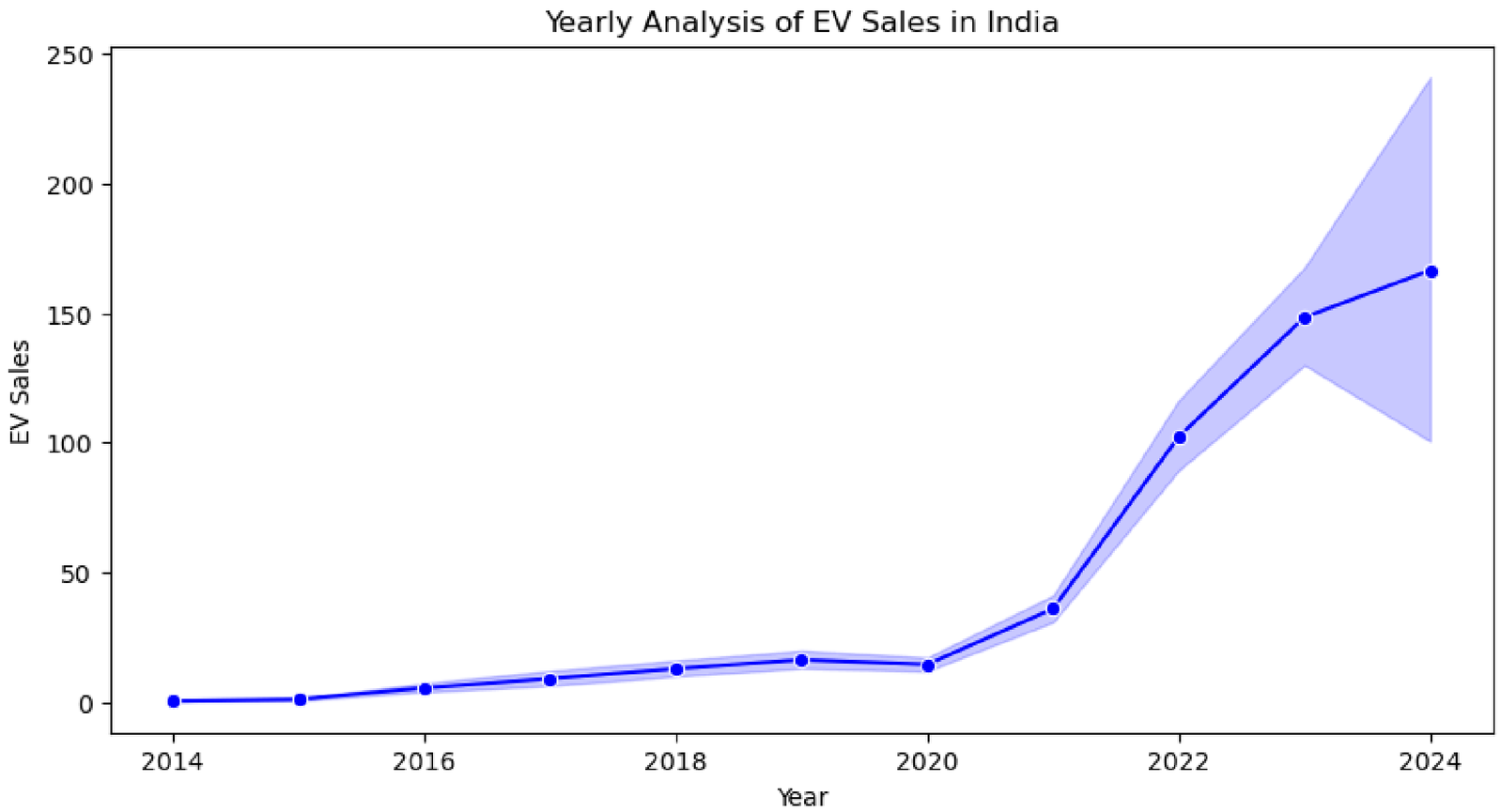
EXPLORATORY DATA ANALYSIS

EDA HELPS IDENTIFY PATTERNS BEFORE BUILDING ML MODELS.

KEY QUESTIONES ANALYSED

- HOW EV SALES EVOLVED OVER TIME?
- WHICH STATES DOMINATE EV ADOPTION?
- WHICH VEHICLE CATEGORIES ARE LEADING?

VISUAL OF EV SALES BY YEAR



YEARLY TREND ANALYSIS

FIELDS

X AXIS → YEAR

Y AXIS → TOTAL EV SALES

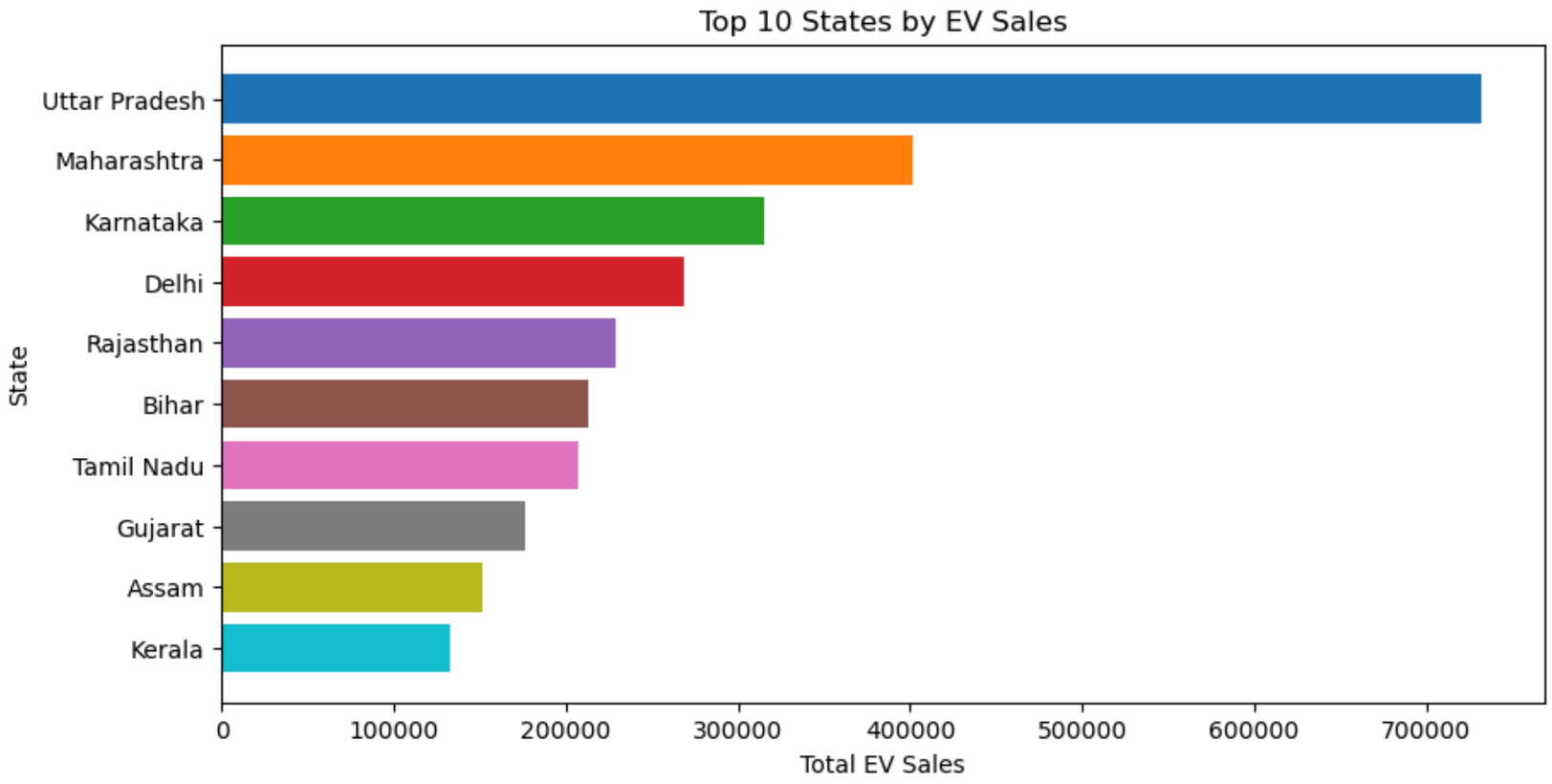
INSIGHT

- Minimal growth before 2019
- Rapid acceleration after 2021
- Indicates strong market adoption

BUSINESS IMPACT

- Confirms EV market expansion phase
- Helps predict future growth trajectory

VISUAL OF EV SALES BY STATE



STATE-WISW TREND ANALYSIS

FIELDS

X AXIS → STATE

Y AXIS → TOTAL EV SALES

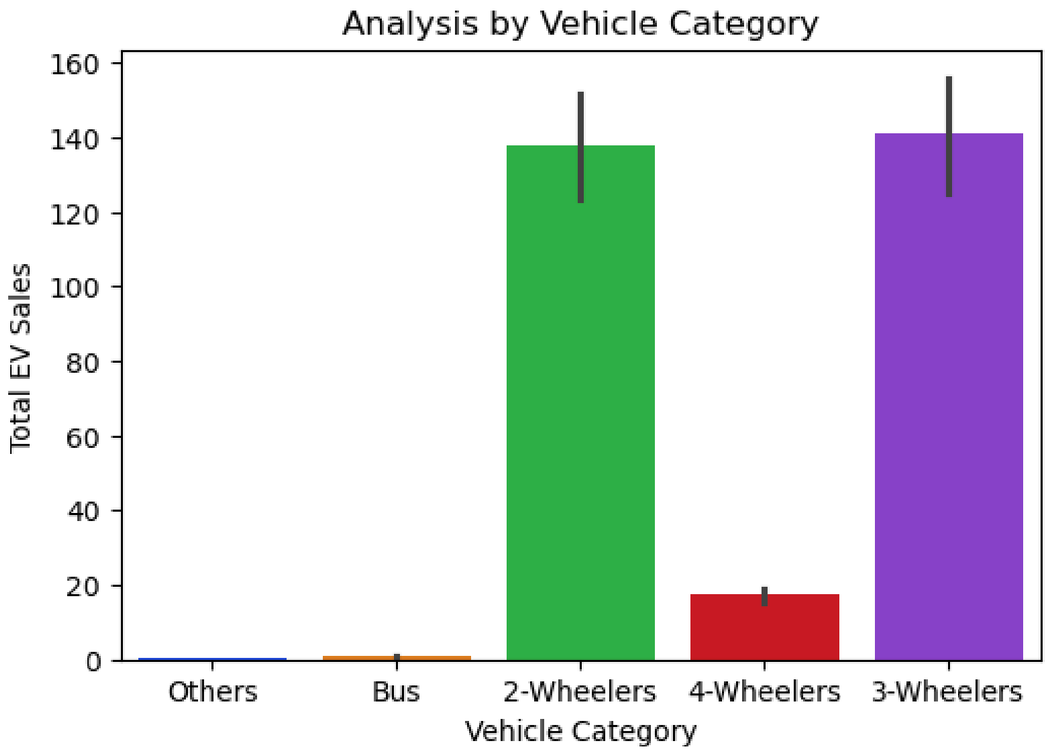
INSIGHT

- Maharashtra, Karnataka, Uttar Pradesh dominate
- Urban and industrial states lead adoption
- Smaller states show lower penetration

BUSINESS IMPACT

- Helps target high-demand regions
- Supports regional policy decisions

VISUAL OF EV SALES BY VEHICLE CATEGORY



STATE-WISW TREND ANALYSIS

FIELDS

X AXIS → VEHICLE CATEGORY

Y AXIS → TOTAL EV SALES

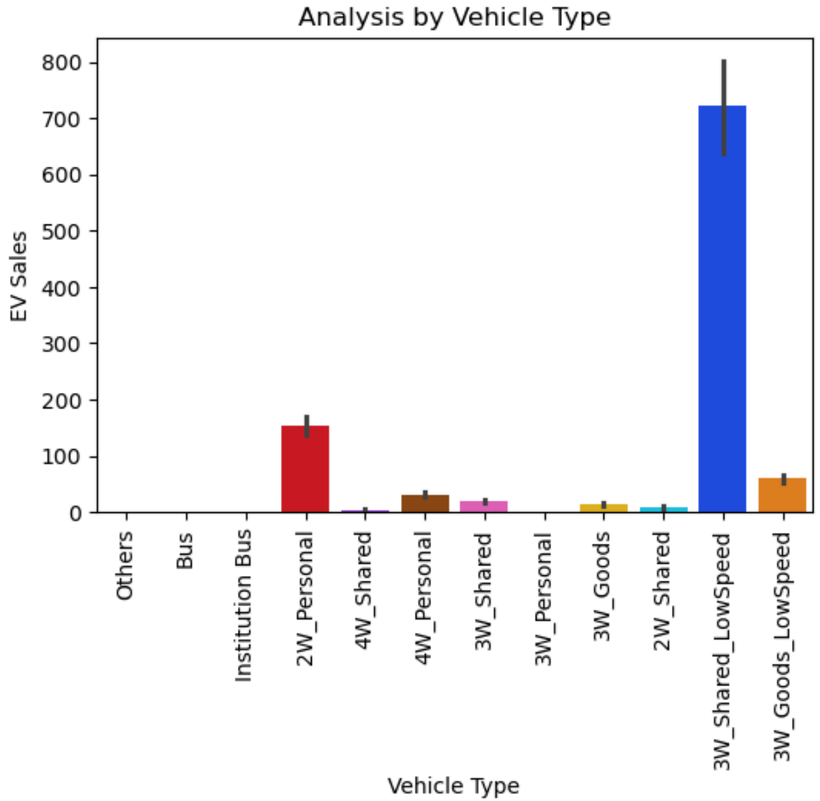
INSIGHT

- 2-Wheelers dominate EV market
- 3-Wheelers show strong adoption in commercial use
- 4-Wheelers adoption still limited

BUSINESS IMPACT

- Focus on affordable EV segments
- Opportunity in passenger EV market

VISUAL OF EV SALES BY VEHICLE TYPE



STATE-WISW TREND ANALYSIS

FIELDS

X AXIS → VEHICLE TYPE

Y AXIS → TOTAL EV SALES

INSIGHT

- Personal EV usage is highest
- Shared mobility also contributes significantly
- Commercial EV segment is growing

BUSINESS IMPACT

- Indicates shift towards sustainable mobility
- Highlights emerging business opportunities

MACHINE LEARNING MODELS

MODELS USED

- Random Forest Regressor
- Gradient Boosting Regressor
- Extra Trees Regressor

WHY THESE MODELS

- Handle non-linear relationships
- Work well with large datasets
- Provide high prediction accuracy

MODEL TRAINING STRATEGY

DATASET SPLIT

- Training Set - 80%
- Testing Set - 20%

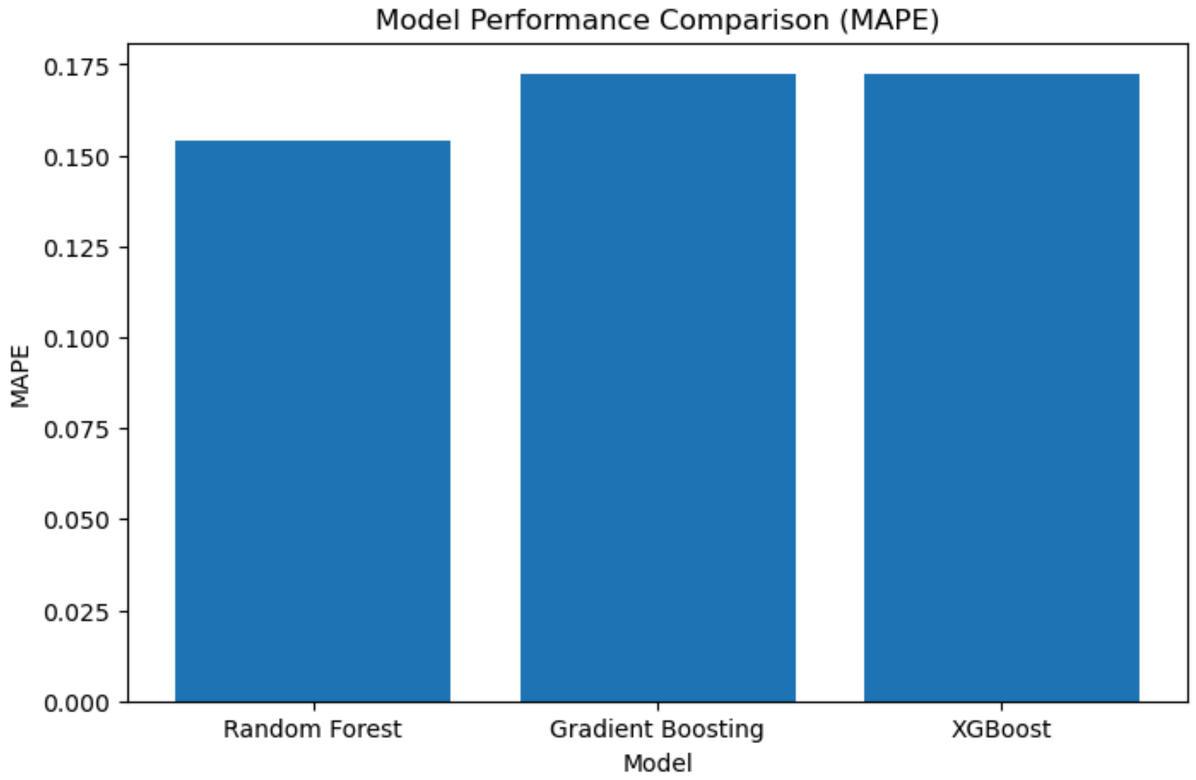
CROSS VALIDATION

TimeSeriesSplit used.

EVALUATION METRICS

- MAE → Error magnitude
- RMSE → Prediction accuracy
- R^2 Score → Model performance

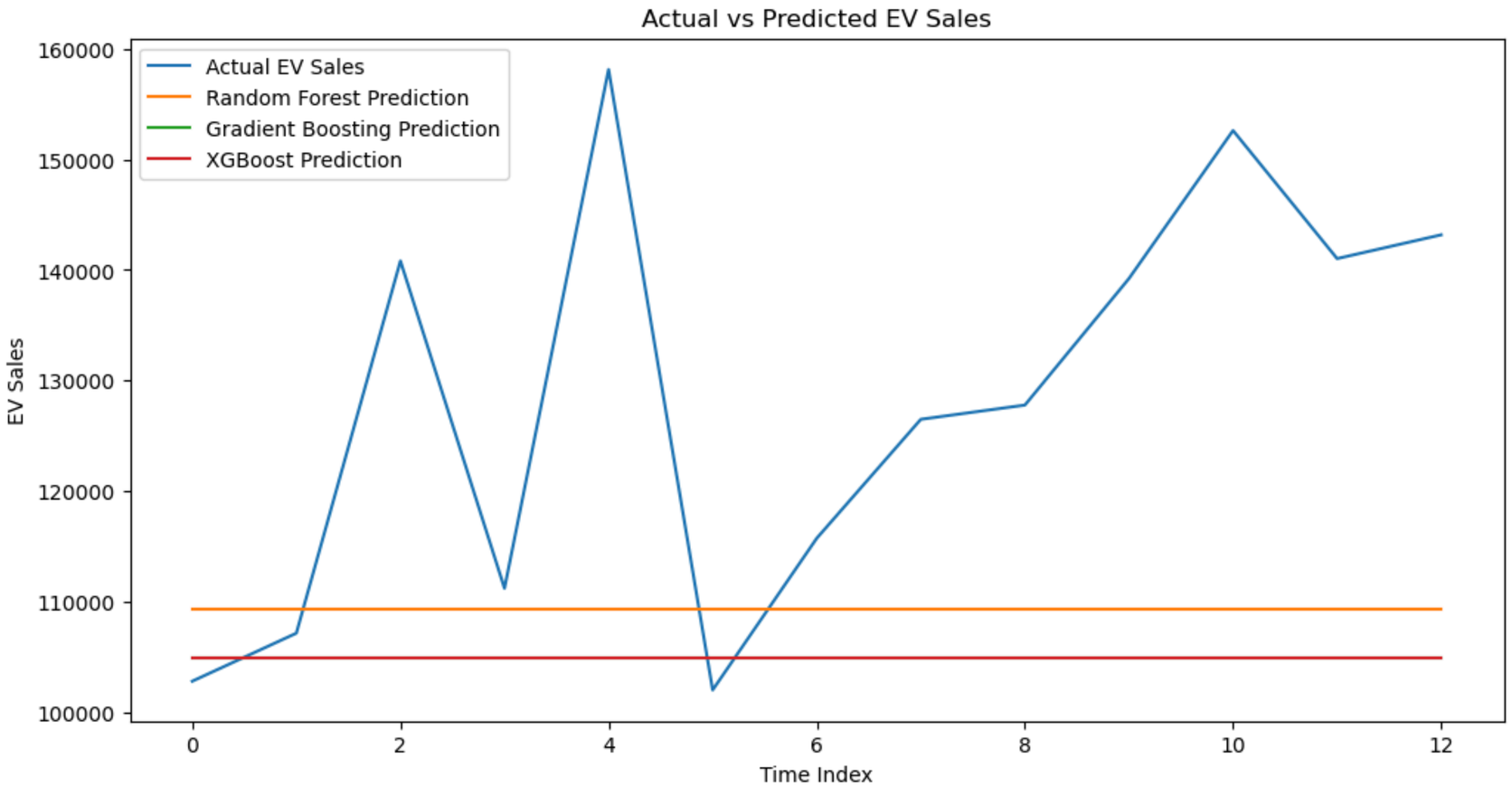
MODEL PERFORMANCE



Conclusion

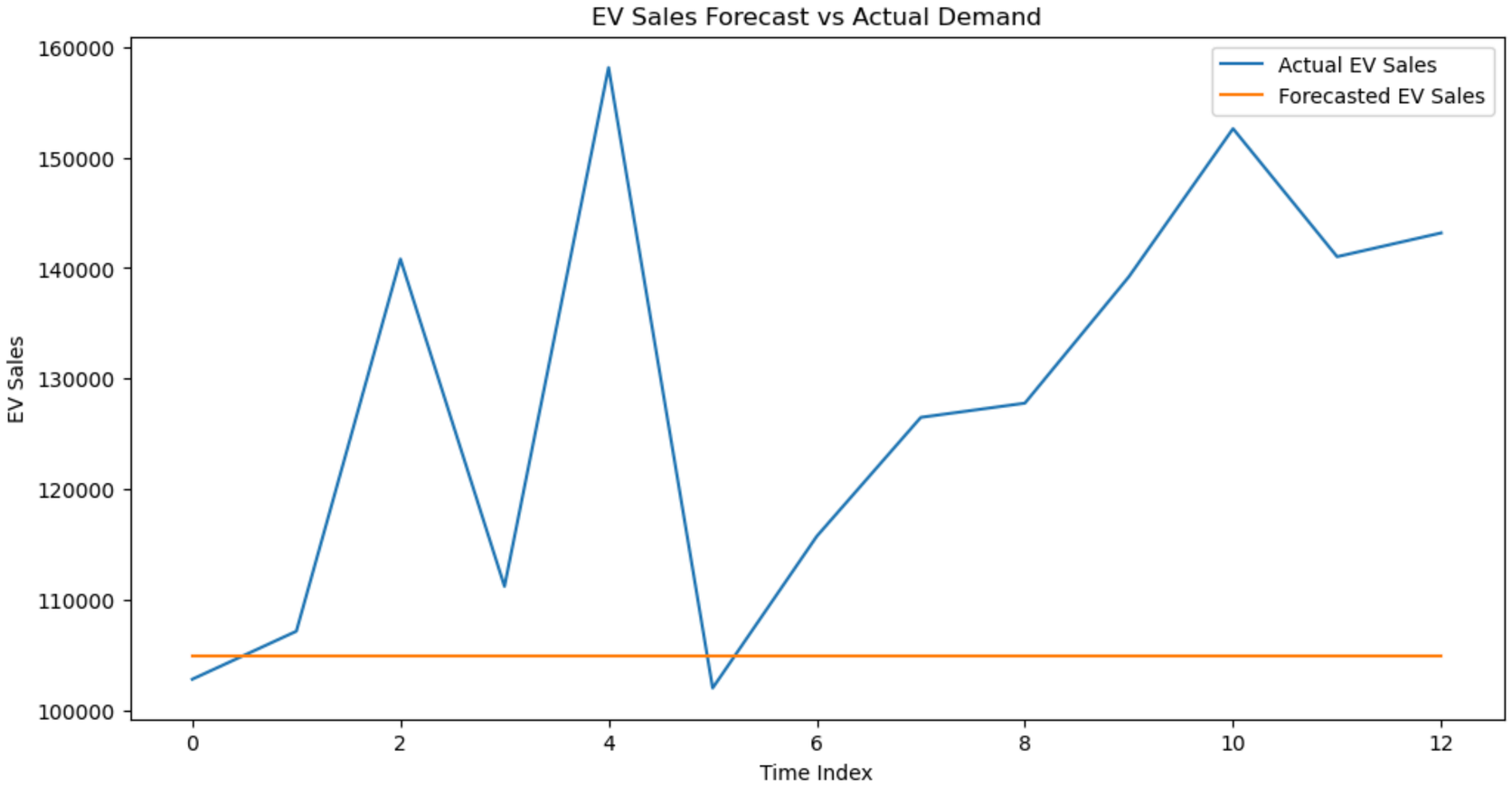
Gradient Boosting Model achieved the best prediction accuracy

PREDICTION VISUALIZATION



Insight: Predictions closely align with actual values

PREDICTION VISUALIZATION



Business Impact: Confirms model reliability for forecasting

FEATURE IMPORTANCE

INSIGHT

- State is the most influential factor
- Vehicle category significantly impacts demand
- Time-based features drive trends

BUSINESS IMPACT

- Helps identify key drivers of EV adoption

FORECASTING

INSIGHT

- EV demand will continue to grow
- Growth expected in emerging states
- Strong upward trend

BUSINESS IMPACT

- Supports long-term planning and investment

ADVANCED MACHINE LEARNING TECHNIQUES

- TIME SERIES RESAMPLING
- CROSS VALIDATION
- PREVENTS OVERFITTING.
- HYPERPARAMETER TUNING
- OPTIMIZES MODEL PERFORMANCE.
- ENSEMBLE MODELING
- COMBINES MULTIPLE MODELS

STRATEGIC INSIGHTS

Insight 1

EV adoption is increasing rapidly

Recommendation

→ Invest in charging infrastructure

Insight 2

Few states dominate the market

Recommendation

→ Expand into underdeveloped regions

Insight 3

2W & 3W dominate EV segment

Recommendation

→ Focus on cost-effective solutions

PROJECT OUTCOME

- Analyzed ~96K EV sales records to identify adoption trends across India
- Discovered state-wise and category-wise demand patterns
- Built machine learning models to predict EV sales accurately
- Identified key drivers of EV adoption such as state and vehicle category
- Developed future demand forecasts to support planning

ADVANCED FEATURES SHOWCASE

- Advanced Analytics & ML Capabilities
- Predictive Modeling & Forecasting Capabilities
- Advanced Feature Showcase
- Machine Learning & Forecasting Highlights
- Data Intelligence & Modeling Capabilities
- AI-Driven Insights & Forecasting
- Core Analytical Capabilities
- Advanced Data Processing & ML Pipeline

BUSINESS IMPACT

- Enables data-driven decision making
- Helps EV companies plan expansion
- Supports government policy planning
- Improves demand forecasting accuracy

PROJECT LIMITATIONS

- External factors not included (fuel price, policy changes)
- Data imbalance across states
- Limited time-series modeling

CONCLUSION

This project demonstrates how data analytics and machine learning can be effectively used to understand and predict EV market trends in India.

- EV market in India is rapidly growing
- Machine learning models effectively predict demand
- Data-driven insights can accelerate EV adoption

“This project not only analyzes EV trends but also provides a predictive framework to support real-world decision making.”

Thank You